

UNIT 6

BLOCKCHAIN TECHNOLOGY



Connect to the topic

1. Look at the photo.
What do you see in the picture?
2. How do you think cryptocurrencies work?
3. How do people store and use cryptocurrencies?
4. How does blockchain relate to cryptocurrency?
5. How is cryptocurrency different from cash?

WARM-UP VIDEO

WATCH AND SPECULATE. Watch the video about blockchain and discuss the questions below.

1. What is blockchain, and how does it work?
2. Why is blockchain secure?
3. How does blockchain remove the need for banks?
4. What information is stored in a blockchain?
5. How are blockchain transactions verified?
6. Why is blockchain seen as a big change?
7. Will blockchain replace banks? Why or why not?



UNIT 6.1 BLOCKCHAIN CORE CONCEPTS

READING AND VOCABULARY

Task 1. READ FOR DETAILS. Read the text and answer the questions below.

What is Blockchain?

Blockchain is technology that allows people to lock data away in a transparent and safe way. Instead of keeping data in one place, like the traditional database, blockchain spreads the information across multiple computers, which are called nodes.

Every piece of data is stored in a block, and a block contains a list of transactions, a timestamp, and a special code called a hash. The hash is created by using a cryptographic algorithm that makes the data in the block special and really hard to alter. If anybody tries to modify even a small piece of data, the hash entirely changes and it's easy to spot any tampering.

Blocks are connected with each other in sequence. If one new block is created, it has the previous block's hash included. This connecting makes the blockchain highly secure. Altering one block is almost impossible unless all the others are altered too, which would require a huge amount of computer power.

Another important idea is that blockchain is decentralized. There is no one server, nor a single company, that holds it. Instead, many people, or nodes, work together to verify and sign off on new blocks. This process is called reaching consensus. Different blockchains use different methods of achieving consensus, like Proof of Work or Proof of Stake, to ensure everything is legitimate.

So what makes blockchain so important? It first creates trust between strangers. In the traditional systems, you do need to have a middleman in the form of, say, a government or a bank, to monitor information and authenticate transactions. Blockchain technology authenticates itself. You can trust the technology rather than a person or company.

Blockchain is also very transparent. Anyone can view the list of transactions by looking at the blockchain. The personal data of the users, however, is protected using cryptographic methods. This feeling of security and transparency allows blockchain not only for digital money, but for supply chains, healthcare, and even online identity management.

In short, blockchain allows for an open, secure, and fair system – and that is the reason why it is growing more important these days.

(adapted from <https://www.ibm.com/think/topics/blockchain>)

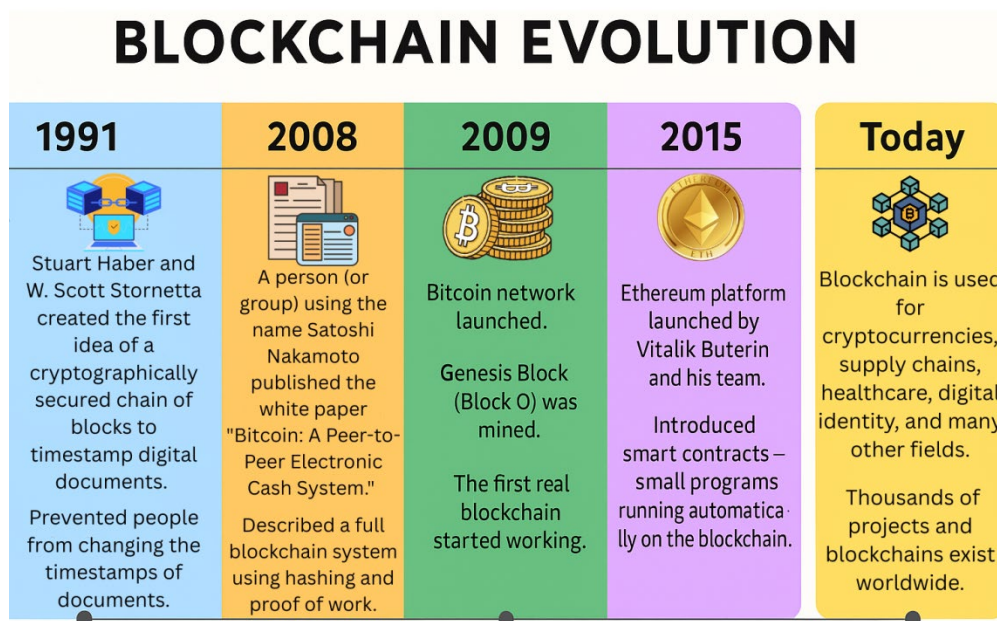
1. What is a blockchain used for?
2. Where is information stored in blockchain technology?
3. What does a block contain? Name two things.
4. What happens if someone changes a small part of a block?
5. What connects the blocks together?
6. Why is it difficult to change information on a blockchain?
7. What does it mean that blockchain is decentralized?
8. What is the name of the process where nodes agree on new blocks?
9. Give one example of a consensus mechanism mentioned in the text.
10. Why can people trust a blockchain without trusting a person?
11. How does blockchain protect personal information?
12. Name two areas (other than cryptocurrencies) where blockchain can be used.

Task 2. WORK WITH WORDS. Match the words from the text to their definitions.

- | | |
|-----------------------|---|
| 1. transparent | a) one computer that is part of a network |
| 2. node | b) not controlled by one single person or organization |
| 3. timestamp | c) when everyone in a group agrees on something |
| 4. hash | d) a special code made from data that helps keep it secure and unique |

5. **cryptographic** e) the act of secretly changing or damaging something
6. **spot** f) to notice or see something
7. **tampering** g) tokens locked in the system to help check transactions and secure the network
8. **decentralized** h) easy to see or understand; nothing is hidden
9. **reach consensus** i) the exact time and date when something happens
10. **stake** j) related to using secret codes to protect information

Task 3. ANALYZE. Study the infographic about blockchain evolution. Answer the questions below.



1. Who are the earliest known contributors to blockchain technology?
2. What was the main goal of the 1991 invention by Haber and Stornetta?
3. What major step did Satoshi Nakamoto take in 2008?
4. What is the Genesis Block, and when was it created?
5. What new feature did Ethereum add to blockchain in 2015?
6. Name two modern uses of blockchain outside of cryptocurrency.

Task 4. WORK WITH WORDS. Fill in the gaps with the words from the box below. There are two extra words you do not need to use.

The Genesis Block

▪ Proof of Work ▪ puzzle ▪ chain of trust ▪ Ethereum ▪ cryptocurrency
▪ transaction fee ▪ hash ▪ cornerstone ▪ mining ▪ broadcast ▪

The Genesis Block, also known as Block 0, was the very first block mined in the Bitcoin blockchain. It introduced key concepts of blockchain 1) _____. In order to add a block to the chain, miners must solve a complex 2) _____, which is part of a mathematical algorithm. It adjusts the difficulty level, depending on how many people are mining at a given time.

When a miner successfully solves the puzzle, they receive a reward – a set amount of 3) _____ – for their work. The new block is then 4) _____ to the network so all users can see it. The process also uses a lot of electricity, especially with 5) _____ systems.

While the Genesis Block was special because it had no 6) _____ before it, all other blocks must link to the hash of the one before them. This 7) _____ is what makes blockchain so secure.

The Genesis Block was mined by Satoshi Nakamoto in January 2009 and contained a hidden message referencing a newspaper headline from that day. This message was likely a comment on the traditional financial system. Even today, the Genesis Block cannot be moved or spent, making it a symbolic 8) _____ of Bitcoin and blockchain history.

Task 5. WORK WITH WORDS. Study the table, then read the sentences and choose the correct mechanism.

Consensus Mechanisms in Blockchain				
Name	How It Works	Used In	Energy Use	Decentralization
Proof of Work (PoW)	Miners solve puzzles using computer power.	Bitcoin, Ethereum (old)	Very high	Very high
Proof of Stake (PoS)	Validators lock coins (stake) to get chosen for validation.	Ethereum, Cardano	Low	High
Proof of Authority (PoA)	A few approved nodes create blocks based on identity or reputation.	Private networks, VeChain	Very low	Low
Delegated PoS (DPoS)	Users vote for a few validators who add blocks.	EOS, TRON	Low	Medium
Proof of Space	Miners use hard drive space instead of processing power.	Chia	Low	Medium
Proof of Burn (PoB)	Coins are permanently destroyed (burned) to earn the right to validate.	Slimcoin, experimental	Medium	Medium

1. This system uses hard drive storage instead of high computer power: _____
2. In this system, a few people are selected by vote to validate blocks: _____
3. Used by Bitcoin, this method needs a lot of electricity: _____
4. Ethereum now uses this low-energy method with locked tokens: _____
5. This system is usually for private companies or governments: _____
6. In this system, coins are destroyed to prove long-term commitment: _____

Task 6. READ FOR DETAILS. Work in A/B/C groups. You will each read a text about how a different Russian government organizations use blockchain technology.

- First, look through your information and use three words from the word box to complete the text.
- Then prepare a short mini-presentation about your information for the other students in your group, using your own words.
- Listen to the other two presentations and ask questions.

recordkeeping ▪ block validation ▪ verified ▪ hash ▪ public access ▪
timestamp ▪ signature ▪ immutability ▪ entry

Group A

Russian Federal Tax Service

The Federal Tax Service of Russia has developed blockchain tools to automate 1) _____ of salary-based loans and income records. This system ensures that all data entries have a valid

timestamp and are linked to a previous block using a unique 2) _____. It reduces fraud, prevents data manipulation, and allows tax offices to cross-check salary declarations without human error. These blockchain systems improve accuracy, speed, and trust in financial 3) _____ between citizens, banks, and the government.

Group B

Rosreestr (Russian Property Registry Service)

Rosreestr has tested blockchain to record land ownership and property transactions. With this system, every change in ownership is stored with a 4) _____, and the chain of records can't be changed without network approval. The result is 5) _____ of legal documents, helping to prevent real estate fraud. When used in a nationwide registry, blockchain makes property data more secure and harder to forge, while still allowing government officials and notaries to access the 6) _____ records.

Group C

Ministry of Health

The Ministry of Health has introduced blockchain for managing vaccination records and medical data. Each new record includes a secure digital 7) _____, which confirms the authenticity of the 8) _____ without exposing private information. Blockchain allows selected hospitals, labs, and authorities to access data quickly while protecting citizens' privacy. The system combines data protection with 9) _____, helping to ensure trust in digital health services and future public health programs.

Task 7. COLLABORATE. Work in pairs or small groups. Match each feature or function of blockchain with the organization(s) that use it: Federal Tax Service, Rosreestr, or Ministry of Health. Some may apply to more than one group.

1. Uses timestamps to track events and ownership changes.
2. Focuses on medical data protection.
3. Increases trust in financial or tax documents.
4. Prevents property fraud with verified records.
5. Involves a digital signature for verification.
6. Operates as part of a national registry.
7. Includes identity verification to restrict access.
8. Protects personal information while keeping public access open.
9. Uses block validation to approve each new entry.
10. Applies blockchain to real-world government services.

WATCHING

Task 8. EXPLORE THE WORDS. Look at the words and study their definitions. Fill in the gaps in the sentences using some of these words.

ledger (n) – a record of transactions or changes kept over time.

batch (n) – a group of items or data processed or stored together.

chain of blocks (phr) – a system where data blocks are linked in time order.

non-destructive (adj) – keeping old data and adding new changes without erasing history.

entry (n) – a single record or line in a database or ledger.

distributed (adj) – shared across multiple locations or computers.

verify (v) – to check or confirm that something is true or correct.

intermediary (n) – a person or organization that acts between two others to build trust.

exposure (n) – the risk of sharing private or sensitive information.

peer-to-peer (adj) – sharing data directly without a central server.

1. Each time someone sends cryptocurrency, a/an _____ is added to the blockchain.
2. In a _____ system, information is stored and managed by many computers instead of just one central server.
3. Blockchain keeps a/an _____, so we can see all past transactions in order.
4. In this system, users connect in a _____ network without going through a bank.
5. We must _____ each new block before adding it to the chain.
6. A blockchain block usually stores a/an _____ of transactions, not just one.
7. One big benefit of blockchain is that it's _____ – no old data is deleted.
8. Using blockchain reduces the need for a/an _____ like a bank or lawyer.
9. A blockchain acts like a/an _____, recording every transaction made across the network.
10. Blockchain reduces _____ to data leaks because personal details are hidden with encryption.

Task 9. WATCH FOR DETAILS. Scan the QR code and watch the video “What is a Blockchain?” Choose the correct answers to the questions.



1. What was blockchain originally created for?
 - A. Tracking land titles
 - B. Powering Bitcoin
 - C. Verifying medical records
 - D. Storing passwords
2. What does the word "blockchain" actually refer to?
 - A. One specific system used by governments
 - B. A type of legal document
 - C. A group of technologies for storing data
 - D. A secure chat platform
3. How is data stored in a blockchain?
 - A. As crypto files
 - B. In a single block
 - C. In a sequence of linked blocks
 - D. On a single server
4. What happens when someone changes data on a blockchain?
 - A. A new block is created with the updated information
 - B. The entire chain is rewritten
 - C. The original data is deleted and new blocks added
 - D. The block is updated and moved to the front
5. What real-world concept is blockchain most similar to?
 - A. A mobile app
 - B. A financial ledger
 - C. A password manager
 - D. An online store
6. Why is blockchain considered non-destructive?
 - A. It never stores personal data
 - B. It works offline and never deletes personal data
 - C. It updates itself every day
 - D. It adds changes without deleting the past
7. What makes blockchain decentralized?
 - A. All users must sign a legal contract

- B. No single computer controls the data
 - C. Only peripheral banks can use it
 - D. It works without central electricity
8. What must happen before a block is added to the chain?
 - A. The user pays a fee
 - B. A lawyer signs off on it
 - C. A cryptographic puzzle must be solved
 - D. The user creates a password and encrypts a block
 9. What is the purpose of "proof of work"?
 - A. To avoid duplicate accounts
 - B. To verify users deleted old blocks
 - C. To prove the user paid a fee
 - D. To confirm the puzzle was solved correctly
 10. How does blockchain reduce the need for intermediaries?
 - A. It connects people directly and verifies the data itself
 - B. It hides ownership details from middlemen
 - C. It uses government-approved contracts
 - D. It transfers information using email

Task 10. COLLABORATE. Work in pairs or small groups. The video mentioned private, public and hybrid blockchains. Discuss how you understand them. Read each situation below. Choose which type of blockchain is the best fit: public, private, or hybrid. Then explain your choice using the useful phrases.

1. A hospital wants to protect patient medical records but share national vaccine data with the public.
2. A global charity wants to make sure all its donation records are open and easy to verify.
3. A large manufacturing company wants to let customers track where a product came from, but it doesn't want to share internal financial data.
4. A city government wants people to see property ownership history, but only city officials should update the data.
5. A private bank wants to share data only among its partner branches and have full control over who can see and edit anything.

Useful Phrases:

- I would choose a hybrid blockchain because it allows...
- A public blockchain is better here...
- A private blockchain works best because...
- The main reason is privacy/security/transparency/efficiency...
- It's important that the public can see it but not change it.

LISTENING

Task 11. LISTEN FOR DETAILS. Scan the QR code and listen to a conversation "Blockchain in Land Registry", where Elena, a government employee at the digital land registry, explains how blockchain works to Tom, a new student intern. Fill in the missing words in the sentences.

1. The digital land registry switched from paper records and private databases to a _____ system.
2. The landowner or notary _____ their documents through the registry portal.
3. A block contains such data as who owns the land, location, size, and a _____.
4. The block also includes a cryptographic _____.



5. Authorized nodes use a _____ protocol to check that the block is valid.
6. After being approved the new blocks are added to the _____.
7. Private details in blockchains like ID numbers are _____.
8. With blockchain technology it became possible to cut costs, reduce _____, and improve trust in public data.

Task 12. LISTEN FOR DETAILS. Scan the QR code and listen to the second part of a conversation “Blockchain in Land Registry” between Elena and Tom. Answer the questions below.



1. Who is allowed to add data to the blockchain system Elena uses?
2. What does Elena do after logging in to the platform?
3. What kind of document is she uploading in this example?
4. What types of information does Elena enter about the property?
Name at least two.
5. What is a hash, and what does it help detect?
6. What happens to the block after it is created and submitted?
7. What do validator nodes check before approving a block?
8. What happens once all nodes agree the data is valid?
9. What kind of information is made public in the blockchain system?
10. What will Tom do tomorrow as part of his internship?

Task 13. WORK WITH WORDS. Match the two parts of the collocation from the listening task.

1. distributed	a) ownership
2. certified	b) signature
3. sync	c) official
4. confirm	d) viewable
5. transfer	e) updates
6. validator	f) across the network
7. digital	g) nodes
8. file	h) ledger
9. registry	i) integrity
10. publicly	j) date

SPEAKING

Task 14. COLLABORATE. Work in pairs. Look at the two columns below. Discuss:

- What are the advantages and disadvantages of each system?
- Which would you prefer for storing land ownership data? Why?

Blockchain vs. Traditional Systems

Traditional Registry	Blockchain-Based Registry
Requires physical documents	Uses digital entries
Involves lawyers/notaries	Peer-to-peer data interaction
Slower approval times	Fast verification via network
Risk of document loss	Stored on distributed ledger
Centralized control	Decentralized and tamper-proof

Useful phrases:

- I think the blockchain version is better because...
- Traditional systems might be safer in case of...
- I would choose _____ for legal reasons/efficiency/data security.

Task 15. COLLABORATE. Work in small groups. Governments are already using blockchain for land ownership. Now imagine other official documents or services that could benefit from this technology. Choose one example from the list (or think of your own). Discuss the four questions below using full sentences. Present your idea to the class if your teacher asks.

Where Else Could Blockchain Help?

- Birth certificates
- Marriage certificates
- University diplomas
- Business licenses
- Voter registration
- Vaccination records
- Driver's licenses
- Building permits

1. Why would blockchain be helpful in this case? (e.g., faster processing, less fraud, no paperwork, etc.)
2. What kind of data would be stored on the blockchain? (e.g., name, date, certificate number, digital signature. etc.)
3. Who should have access to this data? (e.g., government officials, the person themselves, employers. etc.)
4. What risks or challenges might come up? (e.g., privacy issues, digital ID errors, system hacking, etc.)

Task 16. COLLABORATE. Work in groups of 4–6 students. Each student takes a Role Card and stays in character. Place the Situation Cards in a pile, face down. Take turns drawing a card and reading it aloud to the group. As a team, discuss:

- Should blockchain be used?
- Which type: public, private, or hybrid?
- What are the main benefits or risks?

Continue until you've discussed all cards. Be ready to present one solution to the class.

Role Cards

ROLE CARD 1. Ministry of Digital Affairs - You want the government to move to digital systems fast. - You support innovation and better public service. - But you need the system to be easy for everyone to use – even older citizens. - Suggest bold ideas, push for transparency and speed.	ROLE CARD 4. Citizen - You want fast and fair access to services (like ID, licenses, voting). - You hate waiting in long lines or dealing with paperwork. - You're not very tech-savvy and want things to be simple. - Talk about ease of use, mobile apps, and access to your data.
ROLE CARD 2. Legal Advisor - You care about privacy, law, and data rights. - Blockchain must follow all national and international laws. - You're worried about putting personal data on public systems. - Ask questions like: "Who controls the data?", "Can people be identified?"	ROLE CARD 5. Data Security Officer - You focus on keeping public data safe from hackers and leaks. - You like that blockchain prevents tampering. - But open systems might reveal too much if not encrypted. - Ask: "What data is public?" "How is it protected?"
ROLE CARD 3. IT Specialist	ROLE CARD 6. Archivist (Public Records)

- You understand how blockchain works technically. You can explain the difference between public, private, and hybrid chains. - You're concerned about system costs and long-term maintenance. - Share technical benefits but also mention setup challenges.	Office) - You manage historical paper records and scanned documents. - You want a system that preserves records forever. - You're unsure how blockchain can store large files or old documents. - Ask about storage, access to digital copies, and long-term formats.
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Situation Cards

SITUATION CARD 1. Emergency Aid Distribution After a natural disaster, thousands of citizens need quick financial help. The government wants to use blockchain to record and track emergency payments to avoid fraud. Many recipients don't have internet access or digital IDs.	SITUATION CARD 4. Public School Attendance The Education Ministry wants to test blockchain to track student attendance across thousands of public schools – aiming to stop fake reports and improve funding accuracy. Parents are worried it might invade student privacy.
SITUATION CARD 2. Public Infrastructure Projects The Ministry of Construction wants to use blockchain to track large projects – bridges, roads, hospitals – from contract to completion. Citizens demand transparency, but contractors don't want competitors to see their costs.	SITUATION CARD 5. Organ Donation Registry The Health Ministry wants to manage organ donor status using blockchain. Doctors could check status instantly when a transplant is needed. Some citizens worry about ethical misuse or access to personal health info.
SITUATION CARD 3. Court Evidence Logs The justice department is considering blockchain to record every time evidence (like documents or photos) is accessed or changed in a legal case. The data must be secure and time-stamped, but only available to authorized legal teams.	SITUATION CARD 6. State-Owned Art and Museum Collections The Ministry of Culture wants to track every artifact, sculpture, or painting owned by the government using blockchain. It must include ownership history, location, and value but not allow fake copies or digital theft.

Task 17. ANALYZE. Read the case study and discuss possible solutions to the problem.

A group of hospitals and medical research centers joined forces to build a shared system for managing patient data with blockchain. On paper, everything looked smart and could improve security and coordination. But almost as soon as technical planning began, things got complicated. Some institutions liked the idea of a public blockchain, its transparency, the decentralized nature, the built-in traceability. They saw it as a chance to build trust and open up access between hospitals. But not everyone felt the same. A few members were very worried that patient privacy rules wouldn't mix well with such an open system. They preferred a private blockchain with tighter access controls. The other problem is, while the team debated, the demand for a better system only grew. More hospitals were asking when the new platform would be ready, and regulators were increasing pressure to modernize record-sharing systems.

What technical and strategic factors should be considered when choosing between public and private blockchains in this case?

LANGUAGE FOCUS

Task 18. STUDY AND ANALYZE. Look at the rule about relatives, study in what situations they are used.

RELATIVES		
USE	PRONOUN	EXAMPLE
to refer to people (and sometimes animals with personality)	who	Developers who create blockchain applications need strong programming skills.
to refer to things and concepts (and animals without personality)	which	The ledger which records transactions is decentralized.
a formal word for "who" as an object; must be used directly after a preposition	whom	The investor to whom I explained blockchain was very interested.
an informal alternative for who, which, when, where, why (only in defining relative clauses)	that	The platform that supports smart contracts is widely used.
to refer to time	when	The year when Bitcoin was created is 2009.
to refer to place or situation	where	The network where transactions are verified is called a blockchain.
often after "reason"; = "the reason for which" (only in defining relative clauses)	why	Transparency is the reason why many companies adopt blockchain.
the possessive of who and which; can also come after a preposition	whose	A developer whose skills include cryptography is highly valued.
= "the thing(s) which"; only used in defining relative clauses	what	What makes blockchain secure is its encryption.

Defining Relative Clauses	Non-Defining Relative Clauses
These specify which person or thing we are talking about. Removing the clause may cause the sentence to lose meaning. That's the ledger which records all transactions. We <u>can use</u> that instead of which, which makes it more informal. That's the ledger that records all transactions.	These provide extra information. The sentence remains complete without them. Ethereum, which allows smart contracts, is a popular blockchain platform. Satoshi Nakamoto, who created Bitcoin, remains anonymous. We <u>can't use</u> that instead of which, who etc.

Task 19. PRACTICE. Circle the correct word

1. The developer, **who** / **which** created the new blockchain protocol, received an innovation award.
2. Isn't that the exchange **where** / **which** the hacking incident happened last year?
3. The Bitcoin network, **which** / **who** processes thousands of transactions daily, is considered highly secure.
4. There are several reasons **why** / **which** blockchain is improving online payments.
5. This is the platform **where** / **which** I store my cryptocurrency securely.
6. The new engineer in our team, **whose** / **who's** expertise is in smart contracts, is very knowledgeable.

7. Ethereum, **which** / **that** introduced smart contracts, remains one of the most popular blockchain networks.
8. All the developers to **whom** / **who** the security report was sent responded immediately.
9. The year 2009, **when** / **which** Bitcoin was launched, marked the beginning of decentralized finance.
10. Vitalik Buterin, **whose** / **who** contributions to blockchain are widely recognized, co-founded Ethereum.
11. A very secure type of encryption is SHA-256, **which** / **who** is used in Bitcoin mining.
12. The Lightning Network, **which** / **that** enables faster Bitcoin transactions, is gaining adoption.

Task 20. PRACTICE. Write a relative pronoun in each gap. Do not use the word 'that'.

1. The blockchain, _____ stores all the transaction data, is decentralized.
2. I'll never understand the reason _____ they decided to use blockchain for this project.
3. The company _____ developed the new blockchain platform is based in San Francisco.
4. Have you spoken to the person _____ solution could improve data security?
5. Back in 2015, _____ I was learning about cryptocurrencies, no one _____ I knew understood blockchain technology.
6. They've just launched a new app, _____ helps track cryptocurrency transactions, and it's already popular.
7. Bitcoin is a cryptocurrency _____ many people are still learning about.
8. Do you remember the time _____ the blockchain network went down for maintenance?
9. The nodes _____ process transactions in the network are spread across the globe.
10. The developer, _____ algorithm improved the blockchain's speed, was awarded a prize.
11. Peter wants to learn more about blockchain, _____ is becoming increasingly important in many industries.
12. The report _____ you need to read contains details on the latest blockchain developments.
13. _____ makes blockchain so secure is the way each block is linked to the previous one, _____ makes it almost impossible to change past records.
14. Sarah was the blockchain expert _____ everyone relied on.

Task 21. PRACTICE. If a word or phrase in bold is correct, put a tick. If it is incorrect, rewrite it correctly on the line.

1. The blockchain **who we'd implemented it** for security reasons has proven to be very effective.
2. **That** confuses me is why so many people still don't understand how blockchain works.
3. Would any developers **who's** projects involve smart contracts please raise your hands?
4. The year **which** Bitcoin was first created changed the financial world forever.
5. The engineer thought for a second, and then began to write: 'To **Whom** It May Concern...'.
6. The expert **who I spoke to** explained how private keys work.
7. The whitepaper **which I really want to read it** is the one about blockchain.
8. No, the exchange where **you bought your first cryptocurrency there** was shut down last year.
9. Could you just give me one **reason why** you think blockchain is better than traditional databases?
10. The network **where we store transactions on** had a security update last week.
11. The developer **whose wallet you just accessed it** is a cybersecurity expert!

UNIT 6.2 CRYPTOCURRENCIES

READING AND VOCABULARY

Task 22. EXPLORE THE WORDS. Read this short introduction to cryptocurrencies and find words which match the meanings below.

A cryptocurrency is a digital form of money that employs cryptography for the safety of transactions as well as the management of the creation of new units. Unlike bank-issued currencies, cryptocurrencies are decentralised and typically operate on a blockchain – an open book which permanently and publicly accounts for all transactions. The original and most famous cryptocurrency, Bitcoin, came in 2009. It has been joined by thousands more, varying in their attributes and uses. Mining is the process of verifying transactions and adding them to the blockchain, typically receiving new coins as a reward. Many people invest into cryptocurrencies, although their value can be highly volatile. While some people praise cryptocurrencies for improving financial inclusion, others are concerned about their use in criminality and lack of regulation.

1. _____ the practice of securing information using codes.
2. _____ not controlled by a single authority or organization.
3. _____ in a way that cannot be changed or deleted.
4. _____ to put money into something in the hope of gaining a profit.
5. _____ extreme and unpredictable changes in value.
6. _____ the condition of having access to financial services.

Task 23. COLLABORATE. Work in pairs. Choose the best number to complete the sentences about cryptocurrency. Discuss with a partner your point of view. Check the answers with your teacher.

1. The number of cryptocurrencies in existence is around **5,000 / 10,000 / 25,000**.
2. About **10% / 20% / 40%** of people worldwide have used or invested in cryptocurrency.
3. Bitcoin's price dropped by over **50% / 70% / 90%** during its worst bear market.
4. Cryptocurrency mining can use as much electricity as a small **country / city / household**.
5. The total market capitalization of all cryptocurrencies at its peak was over **\$300 billion / \$1 trillion / \$3 trillion**.
6. The average time to confirm a Bitcoin transaction is around **1 minute / 10 minutes / 30 minutes**.
7. The country that first adopted Bitcoin as legal tender was **El Salvador / Switzerland / Singapore**.
8. Ethereum introduced smart contracts in **2013 / 2015 / 2017**.

Which fact surprises you the most? Explain your answer.

Task 24. EXPLORE THE WORDS. Read the sentences and try to understand the bold words from the context. Match the words with their definitions.

1. To stop users from spending the same digital money twice, developers designed systems that detect **double-spending** attempts.
2. **A smart contract** works like a digital vending machine — it automatically does something when certain conditions are met.
3. The platform was built in **layers**, with one part handling user data and another part running programs.
4. **A ring signature** makes it hard to know who really sent a transaction because it mixes the sender's info with others.
5. **A stealth address** is used to hide the recipient's wallet, so no one can trace the payment.
6. High-performance systems that solve complex problems to validate transactions **consume** a lot of electricity.

7. The **underlying** technology behind this system controls how users connect, send money, and stay secure.
8. The network keeps track of all **account values**, including how much digital money each user owns.
 - a) _____ (n) a situation where the same digital asset is used in more than one transaction.
 - b) _____ (n) a private, one-time address that hides who is receiving the money.
 - c) _____ (n) a contract in code that carries out actions automatically.
 - d) _____ (n) the amount of currency and transaction history linked to a digital identity.
 - e) _____ (adj) basic and essential; forming the foundation of something.
 - f) _____ (v) to use up a resource such as energy or time.
 - g) _____ (n) a cryptographic method that hides the sender's identity by mixing their data with others'.
 - h) _____ (n) a separate level or stage in a system, each with its own job.

Task 25. READ FOR DETAILS. Read the text and answer the questions below.

Behind Cryptocurrencies

Cryptocurrencies not only vary in value and popularity but also in technology and use. In the list below, five popular cryptocurrencies are examined and how each excels from a technological standpoint.

Bitcoin (BTC) launched in 2009 as the first decentralized digital currency. Its most significant feature is the proof-of-work algorithm, used to lock up the blockchain and avoid double-spending. It has a peer-to-peer network and a limited 21 million coins available.

Launched in 2015, Ethereum, or ETH, is the source of creating smart contracts; computer programs running on the blockchain automatically when such conditions occur. Therefore, Ethereum is much more than money; it's a decentralized application platform, dApps.

Ripple (XRP) was created for instant and low-value cross-border payments. It does not use mining similar to Bitcoin but is based on a special consensus algorithm that allows transactions to be validated within seconds. Ripple is usually utilized by banks and payment systems.

Cardano, also known by its coin, ADA, is much more academically developed and peer reviewed. It is structured in layers to separate the ledger of account balances from the layer executing the smart contracts. This blockchain uses proof of stake rather than proof of work, taking in much less energy.

Monero (XMR) is anonymity- and privacy-focused. It uses advanced cryptography like ring signatures and stealth addresses to hide transaction details. Monero has the support of people who wish to keep their financial transactions private.

(adapted from <https://cryptohead.io/what-are-the-different-types-of-cryptocurrency/>)

Which cryptocurrency...

1. uses a consensus mechanism other than mining?
2. offers complete anonymity and hides transaction history?
3. is often used to build decentralized apps and services?
4. was the first to introduce a decentralized digital currency?
5. focuses on energy efficiency and scientific design principles?
6. allows instant global payments?
7. is designed with limited supply?
8. supports financial privacy through cryptographic techniques?

- A. Bitcoin
- B. Ethereum
- C. Ripple
- D. Cardano
- E. Monero

9. does not use proof-of-work or proof-of-stake?
10. separates smart contracts from the transaction layer?

Task 26. WORK WITH WORDS. Read the text and fill in the missing words from the box.

Decentralized Apps

intermediary ▪ blockchain ▪ censorship ▪ governance ▪ reward ▪ privacy ▪
smart contracts ▪ open-source ▪ token ▪ Ethereum ▪

Decentralized applications, or dApps, run on 1) _____ networks instead of centralized servers. This means no single company controls the app or its data. dApps are often 2) _____ and resistant to 3) _____, making them popular for finance, gaming, and content sharing. Cryptocurrencies play a key role in dApps. They are used to pay for services, 4) _____ users, or manage transactions automatically through 5) _____. One popular example is Uniswap, a decentralized exchange built on the 6) _____ blockchain. It lets users trade cryptocurrencies directly with each other – without a/an 7) _____. The system is powered by smart contracts and uses its own 8) _____, UNI, to reward participants and support 9) _____. By combining blockchain with cryptocurrency, dApps offer users more control, 10) _____, and transparency.

Task 27. EXPLORE THE IDIOMS. Study the meaning and examples of IT related idioms. Then use them in sentences.

Chain reaction – a series of events triggered by one action.

*The hack caused a **chain reaction** across the crypto network.*

Lock and key – a secure system.

*The data's safe with blockchain's **lock and key** design.*

Pass the buck – to shift responsibility.

*They **passed the buck** when the transaction failed.*

Golden goose – a valuable asset.

*Bitcoin became the **golden goose** of digital currencies.*

Tighten the screws – to increase security or control.

*They **tightened the screws** on blockchain access after the breach.*

In the black – profitable.

*Her wallet was **in the black** after the coin surged.*

Smoke and mirrors – deceptive tactics.

*The startup's promises were **all smoke and mirrors**.*

Fill in the gaps with the correct idiom from the list above. Each idiom is used only once. Sometimes you may need to change the form of the verb.

1. After losing funds, the developers _____ instead of fixing the bug themselves.
2. The glitch in the system set off a _____, crashing wallets one after another.
3. With its perfect encryption, the platform works like a _____, keeping hackers out.
4. Investors cheered as their accounts stayed _____ during the market boom.
5. The new coin was hyped as the next _____, but its value crashed soon after.
6. The scam's wild claims turned out to be just _____, fooling naive traders.
7. After the attack, they _____ on user logins to prevent another breach.

Task 28. COMMUNICATE. Tell the story of a hacker who secretly mines crypto and gets caught because his machines break down. Describe the big mess. Use two or three idioms.

Task 29. WRITE. Write a short, funny story about two friends investing in a hyped-up crypto coin that turns out to be a scam. Use the idioms above.

WATCHING

Task 30. EXPLORE THE WORDS. Match the words from the video to their definitions.

1. hold the balance	a) to figure out or decide
2. equation	b) chance of something happening
3. determine	c) no longer used or useful
4. pending transaction	d) to fix into a system permanently
5. probability	e) in short
6. in a nutshell	f) to keep a record of money in accounts
7. byproduct	g) a payment waiting to be completed
8. self-adjusting	h) changing by itself
9. hard-code	i) side result
10. obsolete	j) a math problem with a missing value

Task 31. WATCH FOR DETAILS. Scan the QR code and watch the video “Bitcoin Mining (Part 1)” to find important figures related to the mining process. Match the sentences with the correct numbers. There are extra figures you do not need to use.



1. The time interval (in minutes) between new blocks being added to the blockchain on average.	6 months
2. The year when mining difficulty started to increase, and people switched to GPU mining.	10
3. The number of CPUs equivalent to the mining power of one GPU.	2013
4. The year when the first ASIC miners appeared.	60
5. The number of people initially mining Bitcoin.	30
6. The range by which FPGAs are faster than CPUs.	2016
7. The year when the pace of miner development started to slow down.	1
8. The frequency at which new ASIC miners were released during the tech race.	3 to 100 times
	2011
	2009
	3
	16 months

Task 32. WATCH FOR DETAILS. Watch the video again. Choose the correct answers.

- What is the main job of a miner in the Bitcoin system?
 - To sell bitcoins to banks
 - To guess a number and update the transaction record
 - To create new types of cryptocurrencies
 - To store other people's passwords
- Why do more powerful computers have a better chance of mining a block?
 - They can guess the answer to the problem more times per second
 - They can connect to banks faster
 - They have better security systems
 - They can create bigger blocks
- What happens after a miner guesses the correct number?

- A. The miner receives help from the bank
 - B. The network deletes the block
 - C. The block is sent to the network for checking
 - D. The miner buys more bitcoins automatically
4. Why did people stop using CPUs for mining?
 - A. CPUs were banned by Bitcoin rules
 - B. CPUs could not connect to the blockchain
 - C. CPUs used too much electricity
 - D. CPUs didn't solve the math problems quickly
 5. What is the main reason mining becomes harder when more people join?
 - A. To keep banks in control
 - B. To make mining more exciting
 - C. To keep the number of new bitcoins steady
 - D. To punish slow computers
 6. Why is the process called "mining" in Bitcoin?
 - A. Because it produces new bitcoins like mining produces gold
 - B. Because miners work under the ground like in real mines
 - C. Because the system needs miners to find gold
 - D. Because miners collect other people's data
 7. What happens if fewer miners are working on the network?
 - A. The mining difficulty increases
 - B. The system creates more complex problems
 - C. It becomes easier to solve the mining problems
 - D. Bitcoin stops working completely
 8. What are ASIC miners special for?
 - A. They can play games and mine at the same time
 - B. They are created only for Bitcoin mining
 - C. They are good for making animations
 - D. They are small and used only by banks

Task 33. WATCH FOR DETAILS. Scan the QR code and watch the video "Bitcoin Mining (Part 2)". Fill in the gaps in the sentences with one or two words you hear.



1. Miners combine their power to form a _____.
2. The mining reward is divided based on each member's _____.
3. Even _____ miners can have a chance to earn Bitcoin through a pool.
4. The number of Bitcoins per block is halved every _____ blocks.
5. The current reward per block is _____ Bitcoins.
6. The hash rate shows how many _____ your computer can make per second.
7. The difficulty of mining depends on how much mining _____ is active.
8. Electricity is used for both powering and _____ the miner.
9. Power consumption is measured in _____.
10. Mining pools usually take a small _____ of your earnings.

Task 34. COMMUNICATE. Work in pairs. Answer the questions below.

1. Why is it difficult for beginners to compete in Bitcoin mining on their own?
2. What happens if a mining pool successfully solves a block?

3. Why might a small miner want to join a mining pool?
4. How many large mining pools exist today, approximately?
5. What was the original Bitcoin block reward in 2009?
6. How often is the Bitcoin block reward halved?
7. What does mining difficulty depend on?
8. Why is electricity cost important when calculating mining profitability?
9. What is the purpose of cooling in Bitcoin mining?
10. Why is it hard to predict the profitability of Bitcoin mining in the future?

LISTENING

Task 35. LISTEN FOR MAIN IDEA. Scan the QR code and listen to a dialogue between Mark, an experienced miner, and Alex who is new in Bitcoin mining. Tick ideas (✓) which were mentioned in the dialogue.



1. Pool fees can affect your profit.
2. Joining a mining pool helps small miners earn more often.
3. Slush Pool has been around for a long time.
4. It's smart to read reviews before picking a pool.
5. Electricity costs are important in mining.
6. Governments regulate which pools you can join.
7. All pools have a fixed 1% fee.

Task 36. LISTEN FOR DETAILS. Listen to the dialogue again. Choose the correct answer for each question.

1. Why is Alex considering joining a mining pool?
 - A. He wants to buy more mining equipment.
 - B. He hasn't earned anything by mining alone.
 - C. He doesn't want to pay electricity bills.
2. What is the usual pool fee mentioned by Mark?
 - A. 5–10%
 - B. 0.5–1%
 - C. 1–2%
3. Why does Mark say solo mining is difficult?
 - A. You need a lot of mining power to compete.
 - B. You have to update the blockchain manually.
 - C. Your computer must be turned off regularly.
4. Which mining pool does Mark currently use?
 - A. Slush Pool
 - B. F2Pool
 - C. Antpool
5. What does Mark say about calculating mining profitability?
 - A. It depends mostly on Bitcoin's market price.
 - B. It's easy if you have the right calculator.
 - C. It involves several changing variables.
6. Why is electricity cost important in mining?
 - A. Because it helps cool the room
 - B. Because miners use a lot of power
 - C. Because it changes daily
7. What do miners need to monitor besides electricity cost?
 - A. Their connection speed
 - B. Their internet provider

- C. Their equipment's power consumption
8. What does Mark say about mining difficulty now?
- A. It's getting easier for new users
- B. It increases as more miners join
- C. It is controlled by pool managers

SPEAKING

Task 37. COLLABORATE. Work in pairs. Use the Internet to understand information about cryptocurrencies. Use the chart to compare cryptocurrencies. Discuss the questions below.

Name	Launch Year	Purpose	Consensus Mechanism	Unique Feature
Polkadot (DOT)	2020	Interoperability between blockchains	Nominated Proof of Stake (NPoS)	Enables multiple blockchains (parachains) to operate together and share data securely.
Solana (SOL)	2020	High-speed dApp and smart contract platform	Proof of History + PoS	Extremely fast processing using Proof of History for time-stamping before validation.
Algorand (ALGO)	2019	Scalable, decentralized applications	Pure Proof of Stake (PPoS)	Instant transaction finality with low fees; no forking thanks to cryptographic sortition.
Tezos (XTZ)	2018	Self-amending blockchain for smart contracts	Liquid Proof of Stake	On-chain governance allows protocol upgrades without forks.
Zcash (ZEC)	2016	Privacy-focused payments	Proof of Work	Offers optional privacy via zero-knowledge proofs (zk-SNARKs)

1. Which one do you think is more suitable for fast transactions?
2. Which one seems more useful for app developers?
3. Which one would you invest in if you were a tech entrepreneur? Why?
4. Which cryptocurrency seems most energy-efficient? Why?
5. Which one do you think is easiest for beginners to understand or use?
6. Which cryptocurrency offers the most innovation in its design or purpose?
7. If you were building a decentralized app, which platform would you choose? Why?
8. Which one do you think is most likely to grow in popularity in the next five years?
9. Would you feel comfortable using any of these for online shopping? Why or why not?
10. Do you think any of these cryptocurrencies could replace traditional banks?

Task 38. COLLABORATE. Work in small groups. Use the Internet to research at least 10 different cryptocurrencies. For each cryptocurrency, find out:

- Can it be mined?
- If yes: What algorithm does it use? What kind of hardware is needed?
- If no: How is it distributed or validated (e.g., staking, pre-mined, delegated proof of stake)?

Consider examples and create a table with the following columns:

Cryptocurrency	Mineable (Yes/No)	Consensus Mechanism	Notes (Mining hardware or staking info)
Bitcoin	Yes	Proof of Work (PoW)	Uses ASIC miners

TRON	No	Delegated PoS	Staking required, SRs validate blocks

Task 39. COLLABORATE. Work in pairs. Match the features to the correct type of crypto wallet.

- | | | |
|---------------------------|---|------------------------------------|
| 1. hot wallet | is connected to the internet for quick access | a) physical device with encryption |
| 2. cold wallet | is stored offline to avoid hacking | b) no electricity or device needed |
| 3. hardware wallet | is USB-like device storing keys offline | c) fast but less secure |
| 4. paper wallet | is when private key written or printed on paper | d) safe for long-term storage |

Task 40. COMMUNICATE. Work in pairs or groups. Answer and explain your preferences using useful phrases.

Would you rather?

- Would you rather use a hot wallet for convenience or a cold wallet for better security?
- Would you rather trust a hardware wallet you carry around or a mobile wallet that's always online?
- Would you rather write your private key on paper or store it in an encrypted app?
- Would you rather have full control of your funds in a non-custodial wallet or let a company manage it in a custodial wallet?
- Would you rather use a wallet with two-factor authentication or one that's simple but only needs a password?
- Would you rather store your savings in a hardware wallet or a paper wallet hidden at home?
- Would you rather use a browser-extension wallet or a desktop-app wallet?
- Would you rather recover your wallet with a 12-word seed phrase or with biometric login?
- Would you rather explain how your wallet works to your grandma or help her set one up?
- Would you rather risk losing access forever with a cold wallet or risk getting hacked with a hot one?

Useful phrases:

- I'd choose... because...
- It's safer/more convenient to...
- In my opinion, the biggest risk is...
- It depends on whether...

Task 41. COLLABORATE. Work in groups of 4 students. Shuffle each deck of cards separately and place face-down. Each player draws one Wallet card and one Situation card for the whole group. Explain why your wallet is high-risk or low-risk in this situation.

Example:

Situation: "Your exchange was hacked."

Student with a Cold Wallet says: "Mine wallet is offline, so I'm safe."

Student with a Hot Wallet may say: "Oh no, I'm at risk, because..."

Crypto Wallet Risk

Wallet Cards

Hot Wallet (mobile/desktop app)	Cold Wallet (offline storage)	Hardware Wallet (USB device)	Paper Wallet (printed key/QR code)
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Situation cards

You need instant access to your funds for a flash sale.	Your laptop gets stolen from a coffee shop.
You lose your hardware wallet device.	A phishing email asks you to “confirm” your wallet password.
You want to move \$10,000 USD worth of crypto tomorrow.	A zero-day bug is discovered in your wallet software.
You spill coffee on your desk and your paper wallet gets wet.	You’re on a business trip in a country with unreliable Wi-Fi.
You forgot your seed phrase.	You accidentally delete the wallet app and haven’t backed up your seed phrase.
Your smartphone’s battery dies right when you need to sign a transaction.	You’re sharing a public computer to check your balance.

Task 42. ANALYZE. Read the case study and discuss possible solutions to the problem.

In 2023, a small business owner in Poland invested in several cryptocurrencies and stored them in a secure digital wallet. He set up two-factor authentication and made sure to write down his seed phrase, the unique backup code everyone says is essential. Months later, something went wrong. His phone reset unexpectedly, and when he went to restore his wallet... no code. He couldn’t find the paper copy anywhere. Desperate, he contacted the wallet provider. But that’s the thing, in decentralized systems, there’s no password reset button. No one has access to your keys except you. And if you lose them, well... that’s it. The coins stay locked up forever. He wasn’t alone. Online forums are full of stories like his. Smart people. Careful people. But one mistake, one missing piece, and their savings are gone.

How can we help people avoid this kind of trap without scaring them away from using these tools in the first place?

LANGUAGE FOCUS

Task 43. STUDY AND ANALYZE. Look at the rule about prepositions, study in what situations they are used.

PREPOSITIONS 1 (VERB+PREPOSITION)

VERBS WITH PREPOSITIONS	EXAMPLE
account for	Crypto transactions must be accounted for in tax reports.
accuse smb of smth	Authorities accuse some platforms of money laundering.
allow for	Smart contracts allow for automated transactions.
benefit from	Investors benefit from early adoption of cryptocurrencies.
blame smb for smth	Some people blame governments for crypto market crashes.
boast about	Some investors boast about their crypto gains.
charge smb for smth	Exchanges charge users for withdrawals.
comply with	Exchanges must comply with anti-money laundering regulations.
consist of	A blockchain consists of blocks that store transaction data.
convert smth into smth	Users can convert Bitcoin into fiat currency.
decide on	They decided on launching their own cryptocurrency.
depend on	Crypto prices depend on market sentiment.
differ from	Cryptocurrencies differ from traditional fiat currencies.
disapprove of	Some traditional banks disapprove of cryptocurrency investments.
distinguish one thing from another	It’s important to distinguish a legitimate token from a scam.

elaborate on	Experts elaborate on blockchain security at conferences.
hedge against	Many investors hedge against inflation by holding Bitcoin.
impose on	Governments impose regulations on crypto exchanges.
insist on	Regulators insist on transparency in cryptocurrency transactions.
lend to	Some DeFi platforms lend crypto to users.
remind of	The volatility of Bitcoin reminds me of stock market crashes.
result from	Market crashes often result from panic selling.
result in	High volatility can result in significant financial losses.
secure smth against smth	Blockchain technology helps secure transactions against fraud.
speculate on	Traders speculate on Bitcoin price movements.
succeed in	Developers who succeed in creating secure wallets attract more users.
trade in	He trades in Bitcoin and Ethereum.
withdraw from	Some investors withdraw from crypto markets during crashes.

Task 44. PRACTICE. Use the verbs from the box in the correct form and the correct preposition.

trade ▪ speculate ▪ succeed ▪ withdraw ▪ insist ▪ depend ▪ benefit ▪
consist ▪ result ▪ blame

1. Crypto traders often _____ Bitcoin, predicting its future value.
2. After years of hard work, the team _____ creating a successful decentralized finance platform.
3. Many investors _____ cryptocurrencies to diversify their portfolios.
4. The CEO _____ using decentralized technologies to solve banking issues.
5. The value of Bitcoin _____ market demand and investor sentiment.
6. They _____ the exchange _____ poor security measures.
7. Some investors _____ the volatility of the crypto market, making huge profits from market swings.
8. He _____ the investment after the market crash.
9. The rise in Bitcoin's popularity _____ an increase in crypto-related scams.
10. The cryptocurrency platform _____ different tools that help users manage their digital assets.

Task 45. PRACTICE. Fill in the correct preposition.

1. The company's blockchain solutions help secure data _____ cyber-attacks.
2. The success of DeFi platforms has resulted _____ better access to finance.
3. Blockchain technology differs _____ traditional databases because it's decentralized.
4. The development of blockchain technology accounts _____ the need for centralized banking systems.
5. Crypto exchanges charge a fee _____ trading digital assets.
6. The company trades _____ cryptocurrencies for investment.
7. To ensure security, you must allow _____ the privacy settings on your crypto wallet.
8. Hackers blame the exchange's weak security _____ the losses.
9. Investors often speculate _____ how regulations affect crypto prices.
10. You should hedge _____ crypto losses during market crashes.
11. Some countries disapprove _____ cryptocurrency regulations to protect investors.
12. They elaborate _____ the new crypto wallet to improve the experience.

Task 46. PRACTICE. Complete the paragraph using the verbs from the box in the correct form. You do not need to use some verbs.

comply ▪ consist ▪ insist ▪ depend ▪ allow ▪ result ▪ elaborate ▪ convert ▪ differ ▪ impose
--

A digital wallet 1) _____ of private and public keys. These keys are needed to send and receive cryptocurrency. The security of a wallet 2) _____ on strong passwords and encryption. Users must 3) _____ with safety rules, like using two-factor authentication, to protect their funds.

Hackers try to steal crypto, which can 4) _____ in big financial losses. To prevent this, developers 5) _____ on better security features, such as fingerprint login. Also, users should never share their private keys, as this could lead to theft.

Some wallets let users 6) _____ one cryptocurrency into another, making trading easier. However, fees and transaction speed 7) _____ from each blockchain network. Choosing a safe wallet is very important for crypto users.

UNIT 6.3 BLOCKCHAIN IN BUSINESS AND BEYOND

READING AND VOCABULARY

Task 47. EXPLORE THE WORDS. Read the sentences and try to understand the bold words from the context. Match the words with their definitions.

1. When the renter didn't reply after receiving the money, it became clear he had **ghosted** the buyer.
2. Thanks to the smart contract, the user received a full **refund** when the file wasn't delivered.
3. A smart contract is a **self-executing** piece of code that performs actions without human help.
4. The contract only runs when a **predefined** condition, such as payment, is met.
5. Smart contracts are often compared to a **vending machine** that gives an item automatically after payment.
6. Once recorded on the blockchain, the data becomes **tamper-proof** and cannot be changed.
7. After a car accident, she filed an **insurance claim** through a blockchain platform.
8. In gaming, people trade **virtual assets** like skins, tokens, or rare items.
9. Blockchain helps track the **authenticity** of goods by confirming their origin.
10. Hospitals can only share patient data if they have the patient's **consent**.
11. The hacker managed to **exploit a flaw** in the smart contract and stole digital funds.
12. Smart contract code must go through **thorough** testing to avoid bugs or errors.

- a) _____ (adj) designed to run automatically when certain conditions are met
- b) _____ (v) to suddenly stop responding or disappear from communication
- c) _____ (n) a request for compensation after loss, damage, or injury
- d) _____ (n) the state of being genuine or real
- e) _____ (n) something you can own in a digital world
- f) _____ (adj) carefully detailed and complete
- g) _____ (adj) already decided in advance
- h) _____ (n) an automatic machine that delivers items after payment
- i) _____ (n) money returned when a service or product is not provided
- j) _____ (adj) protected from being changed or altered
- k) _____ (n) agreement or permission to do something
- l) _____ (v) to take advantage of a weakness in a system

Task 48. READ FOR DETAILS. Read the article. For each sentence, choose the correct ending.

Smart Contracts

Imagine that you reserve an apartment over the internet and pay the advance – and then the landlord simply doesn't respond. No refund, no keys. What if you had an online contract that guaranteed either you get handed over keys upon payment, or you automatically receive a refund? That's the essence of smart contracts.

A smart contract is a self-executing program that exists on a blockchain. It takes specific action as soon as certain pre-set conditions are met. To put it another way, it operates like a vending machine: you insert money, and the machine dispenses a product – without anyone from the store or management being involved.

These digital agreements are referred to as "smart" not because they think, but due to the fact that they possess the ability to verify and enforce the conditions autonomously without the involvement of man. After being deployed, the code cannot be changed, and it makes the process transparent and secure.

Consider a real-life scenario. Alice is willing to sell a digital painting to Bob. They use a smart contract under this condition: "If Bob transfers 2 ETH, send him the artwork." The contract is

uploaded to the Ethereum blockchain. As soon as Bob pays, the program executes automatically and the file is transferred to him. If the payment is not remitted, the contract does nothing. This means both can trust the system – even if they can't trust each other. The blockchain has a permanent, tamper-proof record of the transaction.

Smart contracts are increasingly being applied across many industries:

- Finance: Allowing automatic loan contracts, insurance payouts, and investments.
- Real Estate: Allowing property sales without middlemen or paperwork.
- Gaming: Allowing safe ownership and trade of virtual property.
- Supply Chains: Allowing goods to be tracked across borders and ensuring authenticity.
- Healthcare: Securely and permissioned exchanging patient data between hospitals.

In each of these cases, the main benefit is the same: reducing costs, saving time, and reducing errors through replacing sequential steps with automated, reliable code.

Even though smart contracts reduce human errors, they run the code as written, including whatever bugs it might contain. In 2016, a prominent incident known as The DAO hack involved a hacker exploiting a bug in a smart contract to steal millions of dollars' worth of cryptocurrency. The system ran as written, just not as intended. This is why cautious coding, extensive testing, and using tested templates are required when making smart contracts.

There is no middleman needed with smart contracts. They enable two or more individuals to securely, openly make agreements without banks, lawyers, or notaries. Entrepreneurs and developers view them as opening new ways of creating decentralized services and platforms. With increasingly digital transactions, smart contracts offer a means of creating a more efficient, dependable online world.

(adapted from <https://ethereum.org/en/developers/docs/smart-contracts/>)

1. A smart contract is described as a program that *performs actions automatically when conditions are met / creates human-like responses / controls how websites are accessed.*
2. A vending machine is compared to a smart contract because it *sells products randomly / works without human help / needs someone to supervise the process.*
3. One benefit of using smart contracts is that they *can be edited later / make the process secure and transparent / require lawyers to approve them.*
4. In the example with Alice and Bob, the digital artwork is sent *before the payment is made / only after Bob sends the correct amount / when both users meet in person.*
5. In the finance industry, smart contracts can be used for *insurance claims and investments / building websites / paying bills.*
6. In healthcare, smart contracts can help *transfer patients automatically / share data securely with consent / sell virtual organs.*
7. A big advantage of smart contracts is that they *lower costs and reduce errors / replace blockchain technology / stop all fraud automatically.*
8. The DAO hack showed that smart contracts *can fail if there is a coding flaw / never make mistakes / should be avoided by businesses.*
9. Developers should *write smart contracts by copying others / test them thoroughly and use trusted templates / avoid using them in business apps.*
10. Smart contracts allow developers to *create decentralized platforms / slow down online payments / build private blockchains for banks only.*

Task 49. COMMUNICATE. Work in pairs. Discuss the questions below.

1. Would you trust a smart contract over a human agreement? Why or why not?
2. In what situations would smart contracts be better than traditional contracts? In what situations might they be worse?
3. How do you think smart contracts might change jobs like lawyers or notaries in the future?

4. Should people with no programming knowledge be allowed to create smart contracts? Why or why not?
5. Do you think smart contracts will become part of everyday life (renting, shopping, studying)? Give an example.
6. If someone exploits a bug in a smart contract and gains money, is it stealing or just using the system? Explain your view.
7. Would you use a smart contract to buy something expensive online from a stranger? Why or why not?

Task 50. WORK WITH WORDS. Match business challenges with blockchain solutions.

Business Challenges	Blockchain Solutions
1. Counterfeit luxury goods	a) non-fungible tokens for provenance
2. Slow cross-border payments	b) immutable record of origin
3. Complex multi-party contracts (real estate)	c) permissioned network with access controls
4. Lack of transparency in charity donations	d) smart contracts automate agreements
5. Data errors between departments	e) low-fee token transfers (micropayment tokens)
6. Unauthorized access to sensitive data	f) shared distributed ledger for all donors
7. Fraudulent digital art and collectibles	g) blockchain-based loyalty tokens
8. Inefficient customer loyalty programs	h) single source of truth via distributed ledger

Task 51. WORK WITH WORDS. Read the text and fill in the gaps with the correct words.

Meme Coins

Meme coins are a type of cryptocurrency that often start as jokes or internet trends, but some gain real 1) _____ and popularity. Unlike serious crypto projects with strong technical goals, meme coins are usually created for fun or to build a 2) _____. A well-known example is Dogecoin, which began as a parody of Bitcoin and uses the famous Shiba Inu dog meme. Despite its humorous origin, Dogecoin became widely used for 3) _____ online and even attracted 4) _____ attention.

Technically, meme coins work like other cryptocurrencies – they use blockchain technology, can be 5) _____ or traded, and are stored in digital 6) _____. However, they usually 7) _____ unique technical features and are often more 8) _____. Another example is Trump Coin (\$TRUMP). Associated with President Donald Trump, this coin saw a nearly 60% price increase in April 2025, reaching a market 9) _____ of \$2.7 billion.

Meme coins show how internet culture and community support can influence the success of a cryptocurrency – even without a strong technical 10) _____.

- | | | | |
|------------------|---------------|--------------|-------------|
| 1. A) price | B) value | C) sum | D) cost |
| 2. A) currency | B) company | C) community | D) contract |
| 3. A) swapping | B) stealing | C) boasting | D) tipping |
| 4. A) celebrity | B) private | C) police | D) criminal |
| 5. A) mined | B) hidden | C) converted | D) lost |
| 6. A) wallets | B) purses | C) boxes | D) pockets |
| 7. A) skip | B) waste | C) lack | D) avoid |
| 8. A) efficient | B) simple | C) volatile | D) useful |
| 9. A) cap | B) top | C) limit | D) range |
| 10. A) invention | B) foundation | C) direction | D) program |

Task 52. WORK WITH WORDS. Match each meme coin with the correct fun fact or feature.

1. Dogecoin (\$DOGE)	a) based on a green cartoon frog used in memes
2. Shiba Inu (\$SHIB)	b) a fictional coin representing absurd meme value; often used in classroom games
3. Pepe Coin (\$PEPE)	c) combines cute dog branding with charitable donations and community engagement
4. Baby Doge Coin (\$BABYDOGE)	d) meme coin celebrating “the internet’s favorite actor” with wholesome branding
5. Wojak Coin (\$WOJAK)	e) backed by Melania Trump; gained attention due to political and celebrity branding
6. Milady Meme Coin (\$LADYS)	f) named after Elon Musk’s dog and fueled by celebrity tweets
7. Banana Coin (\$BANANA)	g) uses Viking and doge aesthetics; gained popularity in NFT communities
8. Floki Inu (\$FLOKI)	h) created as a joke but became one of the top cryptocurrencies.
9. Garfield Token (\$GARFIELD)	i) known for its anime-style NFT aesthetic and viral social media support
10. Keanu Coin (\$KEANU)	j) based on a famous orange cartoon cat and trades on its nostalgia value
11. ElonDoge (\$ELONDOGE)	k) inspired by a popular Japanese dog breed and launched as a “Dogecoin killer.”
12. Melania Coin (\$MELANIA)	l) features the “Feels Guy” meme and is part of internet culture symbolism

Task 53. COLLABORATE. Work in pairs. Search for original memes, videos, or images that inspired the meme coins from previous task. Find answers to the questions below.

- What is the original meme?
- Where did it first appear?
- Why do you think it inspired a cryptocurrency?
- Is it funny, ironic, nostalgic, or something else?

Show the meme to the group and explain its connection to internet culture and digital communities.

Task 54. COMMUNICATE. Work in pairs. Discuss the questions below.

1. Do you think memes are a good basis for creating digital money?
2. Can meme coins be taken seriously, or are they just a joke?
3. Which meme coin would you invest in and why?
4. What makes a meme go viral?
5. How do communities influence the popularity of a coin?

Task 55. COLLABORATE. Work in groups or pairs. Invent your own fictional meme coin. You must:

- Name it
- Design the meme or visual
- Choose its symbol (e.g., \$LOL, \$GRUMPY)
- Explain its “story,” purpose, or use.

WATCHING

Task 56. EXPLORE THE WORDS. These sentences will help you understand the video about NFTs. Guess the meaning of each bold word and choose the best meaning.

1. Susan was afraid someone might **replicate** her digital paintings without permission.
A) erase B) copy exactly C) change the colors D) buy quickly
2. NFTs are easily **transferable**, allowing owners to sell them online.
A) unable to be moved B) available for printing C) easy to give or sell to others D) possible to delete
3. An NFT is linked to a **digital asset**, like music, video, or artwork.
A) a physical object B) a virtual item with value C) a banknote D) a printed photo
4. Unlike NFTs, Bitcoin is **fungible** – you can exchange one for another of the same value.
A) not valuable B) hard to store C) exchangeable and identical in value D) slow to trade
5. Non-fungible items like paintings cannot be created **in bulk**.
A) with colors B) by accident C) in large quantities D) in secret
6. Even if a copy looks similar, it is not **authentic** without the NFT.
A) original and real B) fake C) beautiful D) legal
7. Each NFT contains **distinguishable** details, such as the owner's name and transaction history.
A) impossible to understand B) very expensive C) illegal D) easily told apart
8. CryptoPunks lets users buy and store 10,000 digital **collectibles**.
A) stolen artworks B) types of money C) outdated technology D) items collected and valued by people
9. NFTs help protect digital art from **forgery**.
A) storage B) deleting data C) copying without permission D) selling online
10. For Susan, NFTs were a **boon** because they protected her art and boosted sales.
A) problem B) mistake C) law D) benefit

Task 57. WATCH FOR MAIN IDEA. Scan the QR code and watch the video “Non Fungible Token (NFT)”. Answer the question which appears at the end of the video.

What makes any item non-fungible?

- A. Exchangeability
- B. Unique digital signature.
- C. Tradability.
- D. Distributed public ledger.



Task 58. WATCH FOR DETAILS. Watch the video again. Answer the questions below.

1. Why was Susan worried about holding her exhibition online?
2. What solution did Mark suggest to protect Susan's artwork?
3. How does an NFT define the ownership of a digital asset?
4. Why is the painting of the Last Supper considered non-fungible?
5. What type of information does each NFT contain?
6. What technology is used to create NFTs?
7. Which blockchain platform do NFTs primarily exist on?
8. How are NFTs different from cryptocurrencies like Bitcoin or Ethereum?
9. What is CryptoPunks?
10. What was the result of Susan using NFTs for her artwork?

LISTENING

Task 59. EXPLORE THE WORDS. Look at the words and study their definitions. Fill in the gaps in the sentences using some of these words.

1. During the **election**, millions of citizens choose their next president.
2. When you arrive at the polling station, you receive a paper **ballot** with all the candidates' names.
3. The old voting system had a single server, a major **point of failure**, so when it crashed, no one could vote.
4. The government rejected that voting app because it ran on **proprietary** software and citizens couldn't inspect its code.
5. The committee's decision process was too **opaque**, so no one knew how they reached their conclusions.
6. Only **eligible** residents, those over 18 and registered, may participate in the referendum.
7. After logging in, you click "Submit" to **cast a vote** for your preferred option.
8. A certified **auditor** reviewed the ballot counts to ensure there were no mistakes.
9. The pilot was **small-scale**, involving only one district instead of the entire state.
10. A brief **network outage** delayed vote submissions until connectivity returned.
11. Because the old system had weak security, there was a report of ballot **theft** during the primary.
12. The new voting app runs on a **permissioned** blockchain, so only approved nodes can add vote records.

- a) _____ a system crash or weakness that stops everything working
- b) _____ someone qualified by law or rules to do something
- c) _____ a sudden loss of internet or server connection
- d) _____ run on a limited or reduced scope
- e) _____ a paper or digital form used to record a choice in a vote
- f) _____ a formal process to choose a person for public office
- g) _____ to officially send in your choice in an election
- h) _____ software whose source code is hidden from users
- i) _____ not clear or transparent
- j) _____ a professional who checks and verifies financial or procedural records
- k) _____ the illegal taking of something
- l) _____ a controlled-access network where only approved participants can write data

Task 60. LISTEN FOR DETAILS. Scan the QR code and listen to the audio "Blockchain Voting". You will hear an interview with Dr. Petros, an expert in secure digital elections. Fill in the missing words in the sentences.

1. Traditional elections often use paper ballots or _____ digital systems.
2. Paper ballots can be _____, damaged, or changed.
3. Blockchain voting uses a _____ ledger shared among approved participants.
4. Eligible voters log in with a secure digital _____ before casting a vote.
5. Each vote is added as an _____ transaction, ensuring it cannot be changed.
6. No _____ information is stored on-chain, which protects voter identity.
7. Auditors can _____ the total count without knowing who voted for whom.
8. Estonia first introduced online voting in the year _____.
9. In 2018, West Virginia held a mobile blockchain voting _____ for military voters.



10. Some governments are testing blockchain in limited _____ elections.

Task 61. WORK WITH WORDS. Match the two columns to make collocations from the interview above.

1. paper	a) trust
2. tamper	b) theft
3. proprietary	c) privacy
4. public	d) ballot
5. independent	e) pilots
6. voter	f) nodes
7. unique	g) elections
8. real-world	h) with results
9. local	i) token ID
10. device	j) software

Task 62. COMMUNICATE. Work in pairs. Discuss the questions below.

1. Why do some people think blockchain could improve elections?
2. What are the main problems with traditional voting methods that blockchain tries to solve?
3. How does blockchain protect both vote accuracy and voter privacy at the same time?
4. Do you think everyone would trust a voting system if they couldn't see their personal vote recorded? Why or why not?
5. What risks still exist with using blockchain for voting?
6. Would you feel more confident voting online if blockchain technology was used? Why or why not?
7. Should blockchain voting be used only for small-scale elections at first, or is it ready for national elections? Explain.

SPEAKING

Task 63. COLLABORATE. Work in small groups. Read and analyze the cases. Answer the questions below.

- What problem was the company facing?
- How did blockchain solve it (which features)?
- What were the main benefits and any drawbacks?
- Would you recommend this solution to a competitor? Why/why not?

Case Study 1

Maersk's TradeLens Shipping Platform

Maersk, the world's largest container shipping company, faced delays, paperwork errors, and lack of visibility across global supply chains. Documents passed through multiple intermediaries, causing days – or even weeks – of waiting and extra costs. Maersk partnered with IBM to launch TradeLens, a distributed ledger platform. Each participant (shippers, ports, customs authorities, freight forwarders) runs a node on the network. Every container's journey is recorded as immutable blocks: timestamped events (loading, unloading, customs clearance), shared document library, smart contracts to trigger automatic notifications or payments once delivery is confirmed.

Case Study 2

De Beers' Diamond Provenance Tracking

The diamond industry has long struggled with "conflict diamonds" – stones mined in war zones, used to finance armed conflict. Buyers and governments want proof that diamonds are "clean." De Beers developed Tracr, a private, permissioned blockchain. At each step (mining, cutting, polishing,

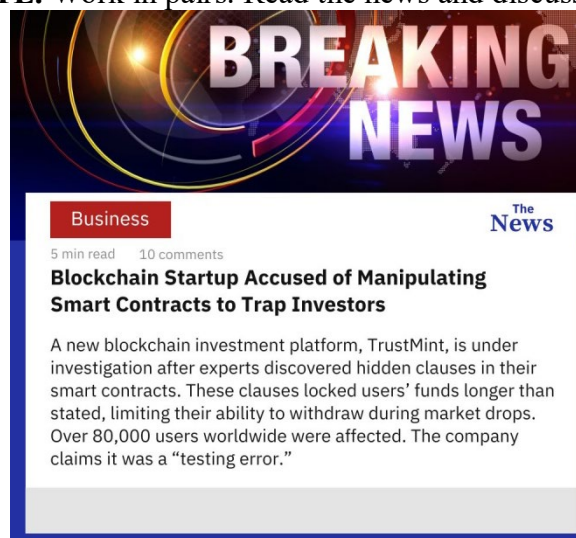
distribution) each diamond gets a digital ID (its “digital twin”). Key features: immutable record (each ownership change and grading report is permanently stored), QR code proof (customers scan jewelry to see its history), access control (only approved companies can add data; everyone can read).

Case Study 3

IBM Food Trust for Supply-Chain Transparency

Food recalls due to contamination (like salmonella) cost billions and risk public health. Trace-back investigations, identifying the farm, processing plant, or shipment, take days or weeks. IBM Food Trust on Hyperledger Fabric connects farmers, processors, distributors, and retailers on a permissioned blockchain. Each batch of food has a lot number, and every step is recorded on a private blockchain. Features include: end-to-end traceability(farm → warehouse → store shelves), smart alerts (if one batch is unsafe, the system flags all related batches), shared data platform (regulators see the same secure information).

Task 64. COMMUNICATE. Work in pairs. Read the news and discuss the questions below.



1. What was the public's reaction to the smart contract scandal?
2. What legal actions might be taken against the startup?
3. How could this incident affect trust in blockchain-based financial tools?
4. Should smart contracts always be audited by third parties before launch?
5. Who should cover the losses: the platform, developers, or the users?



1. How has the public responded to the failure of the blockchain voting system?

2. What steps should election officials and developers take to investigate and prevent future errors?
3. How can blockchain failures influence public trust in digital democracy tools?
4. Who should take responsibility: the government, the developers, or regional tech support teams?
5. Should voting ever be fully digital and blockchain-based, or are some risks too great?

Task 65. COLLABORATE. Divide into teams with 3-5 players. Listen to a trivia question read by a teacher. You have 30 seconds to discuss and give the answer. A correct answer earns 1 point.

Blockchain and Crypto Trivia Game

1. What kind of blockchain allows only approved participants to write data?
2. True or False: Cryptocurrency transactions are usually anonymous but traceable.
3. True or False: A blockchain ledger can be edited by any user at any time.
4. What is the name for the file that stores your cryptocurrency keys securely?
5. What is the process of earning cryptocurrency by solving math problems?
6. What kind of coin is built on top of an existing blockchain, not its own?
7. True or False: Smart contracts require a lawyer to be enforced.
8. True or False: A miner can always update two blocks in a row.
9. What type of digital agreement runs automatically when certain conditions are met?
10. True or False: All cryptocurrencies are based on the Bitcoin blockchain.
11. What kind of data structure is used in blockchain to link information in order?
12. What term describes cryptocurrency tokens that are not interchangeable?
13. What technology is used to ensure transactions can't be changed or faked?
14. What is the digital ID or proof attached to a piece of art in an NFT?
15. What kind of wallet is stored offline and considered very secure?
16. True or False: NFTs can represent digital art, music, and even game items.
17. What company partnered with IBM to create a blockchain for global shipping?
18. What is the term for the part of the blockchain where each entry is recorded?

WRITING

Task 66. WRITE. Imagine you are a student startup founder or IT project manager. Your team has developed a blockchain-based product or service (you can choose from the ideas below). Write a short business proposal (100–130 words) to a potential client. Choose one idea to propose:

- A blockchain system for document verification (e.g. academic certificates, IDs)
- A smart contract platform for automating supplier payments
- A blockchain-based voting system for university elections or company boards
- A crypto wallet plugin for secure in-app purchases in gaming platforms
- A blockchain traceability tool for food or fashion industry (anti-counterfeit)

Your proposal should include:

- A brief introduction of you or your team
- The problem your product solves
- Key benefits and features of your solution
- A trial proposal or meeting offer

Useful phrases:

- We would like to propose...
- Our system provides...
- This solution helps reduce/increase/improve...
- We suggest starting with a pilot project...
- Please find attached...

- We would be happy to arrange a meeting to discuss this in more detail.
- Looking forward to your reply.

LANGUAGE FOCUS

Task 67. STUDY AND ANALYZE. Look at the rule about nouns and adjectives with prepositions, study in what situations they are used.

PREPOSITIONS 2 (NOUN+PREPOSITION, ADJECTIVE+PREPOSITION)

NOUNS WITH PREPOSITIONS	EXAMPLE
attack on	An attack on the blockchain network can compromise security.
ban on	Some countries have a ban on cryptocurrency trading.
basis for	Blockchain technology is the basis for decentralized finance.
cap on	Some cryptocurrencies have a cap on the total number of coins that can be mined.
contribution to	Developers make contributions to open-source blockchain projects.
control over	Decentralized exchanges give users more control over their funds.
demand for	There is a growing demand for secure crypto storage solutions.
difficulty in	There is difficulty in mining Bitcoin due to high competition.
impact on	The impact on energy consumption is a major concern for crypto mining.
increase in	The increase in transaction fees has affected many users.
lack of	A lack of regulation can lead to increased fraud in the crypto space.
protection against	Encryption provides protection against hacking attempts.
restrictions on	There are restrictions on crypto mining in certain regions.
threat to	Phishing scams are a major threat to crypto wallets.
ADJECTIVES WITH PREPOSITIONS	EXAMPLE
afraid of	Some investors are afraid of the risks involved in DeFi platforms.
alert to	Users should always be alert to phishing scams targeting crypto wallets.
available for	This crypto wallet is available for both desktop and mobile users.
busy with	Miners are busy with upgrading their hardware.
careful with	Be careful with private keys; losing them means losing access to your funds.
eager for	Many investors are eager for new opportunities in blockchain technology.
efficient at	Some blockchains are more efficient at processing transactions than others.
essential for	Secure key storage is essential for protecting digital wallets.
limited to	Some blockchain networks are limited to a fixed number of validators.
responsible for	Miners are responsible for securing the blockchain through computations.
slow at	Older blockchains are slow at processing large volumes of transactions.
vulnerable to	Online crypto wallets can be vulnerable to hacking attacks.

Task 68. PRACTICE. Use the nouns and adjectives from the box and the correct preposition.

demand ▪ limited ▪ restrictions ▪ afraid ▪ increase ▪ difficulty ▪ ban ▪
contribution ▪ eager ▪ efficient

1. The _____ cryptocurrency transactions in some countries makes trading difficult.
2. There is a growing _____ secure blockchain networks due to the rise of cyberattacks.
3. Miners are _____ using advanced hardware to improve their performance.
4. Many new investors are _____ new trends in blockchain technology, hoping to profit.
5. Some blockchain systems are _____ large transactions due to their low capacity.
6. There are _____ the amount of cryptocurrency that can be transferred between countries.
7. Developers made a significant _____ the success of the new blockchain protocol.
8. Users are _____ losing access to their funds due to forgotten private keys.
9. Many users face _____ understanding how smart contracts work.
10. There has been a/an _____ the number of hackers targeting crypto exchanges.

Task 69. PRACTICE. Choose the correct variant.

1. The team is _____ upgrading the blockchain protocol to improve security.
A. busy with
B. vulnerable to
C. careful with
2. The new regulations include a _____ the number of Bitcoin transactions per minute.
A. cap on
B. attack on
C. basis for
3. Blockchain is the _____ modern financial system.
A. threat to
B. basis for
C. ban on
4. Cryptocurrency investors are _____ hacking attempts and phishing scams.
A. essential for
B. alert to
C. slow at
5. A _____ communication between wallet users and blockchain validators is critical to security.
A. impact on
B. lack of
C. control over
6. Cryptocurrencies are _____ cyberattacks, especially when stored in online wallets.
A) efficient at
B) careful with
C) vulnerable to
7. Users need to have _____ their private keys to avoid unauthorized access to their wallets.
A. control over
B. protection against
C. attack on
8. Encryption algorithms provide _____ unauthorized access to crypto transactions.
A. cap on

- B. protection against
 - C. lack of
9. This mobile wallet is _____ both Android and iOS users.
- A. available for
 - B. busy with
 - C. vulnerable to
10. Investors are _____ new trends in cryptocurrency, hoping to make high returns.
- A. slow at
 - B. afraid of
 - C. eager for